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on kids and learning

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before i had kids, i was afraid of having kids. up to that point i felt about kids the way the young augustine felt about living virtuously. i'd have been sad to think i'd never have children. but did i want them now? no.

if i had kids, i'd become a parent, and parents, as i'd known since i was a kid, were uncool. they were dull and responsible and had no fun. and while it's not surprising that kids would believe that, to be honest i hadn't seen much as an adult to change my mind. whenever i'd noticed parents with kids, the kids seemed to be terrors, and the parents pathetic harried creatures, even when they prevailed.

when people had babies, i congratulated them enthusiastically, because that seemed to be what one did. but i didn't feel it at all. "better you than me," i was thinking.

now when people have babies i congratulate them enthusiastically and i mean it. especially the first one. i feel like they just got the best gift in the world.

what changed, of course, is that i had kids. something i dreaded turned out to be wonderful.

partly, and i won't deny it, this is because of serious chemical changes that happened almost instantly when our first child was born. it was like someone flipped a switch. i suddenly felt protective not just toward our child, but toward all children. as i was driving my wife and new son home from the hospital, i approached a crosswalk full of pedestrians, and i found myself thinking "i have to be really careful of all these people. every one of them is someone's child!"

so to some extent you can't trust me when i say having kids is great. to some extent i'm like a religious cultist telling you that you'll be happy if you join the cult too — but only because joining the cult will alter your mind in a way that will make you happy to be a cult member.

but not entirely. there were some things about having kids that i clearly got wrong before i had them.

for example, there was a huge amount of selection bias in my observations of parents and children. some parents may have noticed that i wrote "whenever i'd noticed parents with kids." of course the times i noticed kids were when things were going wrong. i only noticed them when they made noise. and where was i when i noticed them? ordinarily i never went to places with kids, so the only times i encountered them were in shared bottlenecks like airplanes. which is not exactly a representative sample. flying with a toddler is something very few parents enjoy.

what i didn't notice, because they tend to be much quieter, were all the great moments parents had with kids. people don't talk about these much — the magic is hard to put into words, and all other parents know about them anyway — but one of the great things about having kids is that there are so many times when you feel there is nowhere else you'd rather be, and nothing else you'd rather be doing. you don't have to be doing anything special. you could just be going somewhere together, or putting them to bed, or pushing them on the swings at the park. but you wouldn't trade these moments for anything. one doesn't tend to associate kids with peace, but that's what you feel. you don't need to look any further than where you are right now.

before i had kids, i had moments of this kind of peace, but they were rarer. with kids it can happen several times a day.

my other source of data about kids was my own childhood, and that was similarly misleading. i was pretty bad, and was always in trouble for something or other. so it seemed to me that parenthood was essentially law enforcement. i didn't realize there were good times too.

i had plenty of happy times before i had kids. but if i count up happy moments, not just potential happiness but actual happy moments, there are more after kids than before. now i practically have it on tap, almost any bedtime.

people's experiences as parents vary a lot, and i know i've been lucky. but i think the worries i had before having kids must be pretty common, and judging by other parents' faces when they see their kids, so must the happiness that kids bring.

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notes

1 adults are sophisticated enough to see 2 year olds for the fascinatingly complex characters they are, whereas to most 6 year olds, 2 year olds are just defective 6 year olds.

thanks to trevor blackwell, jessica livingston, and robert morris for reading drafts of this.

adults lie constantly to kids. i'm not saying we should stop, but i think we should at least examine which lies we tell and why.

there may also be a benefit to us. we were all lied to as kids, and some of the lies we were told still affect us. so by studying the ways adults lie to kids, we may be able to clear our heads of lies we were told.

i'm using the word "lie" in a very general sense: not just overt falsehoods, but also all the more subtle ways we mislead kids. though "lie" has negative connotations, i don't mean to suggest we should never do this—just that we should pay attention when we do. [1]

one of the most remarkable things about the way we lie to kids is how broad the conspiracy is. all adults know what their culture lies to kids about: they're the questions you answer "ask your parents." if a kid asked who won the world series in 1982 or what the atomic weight of carbon was, you could just tell him. but if a kid asks you "is there a god?" or "what's a prostitute?" you'll probably say "ask your parents."

since we all agree, kids see few cracks in the view of the world presented to them. the biggest disagreements are between parents and schools, but even those are small. schools are careful what they say about controversial topics, and if they do contradict what parents want their kids to believe, parents either pressure the school into keeping quiet or move their kids to a new school.

the conspiracy is so thorough that most kids who discover it do so only by discovering internal contradictions in what they're told. it can be traumatic for the ones who wake up during the operation. here's what happened to einstein:

> through the reading of popular scientific books i soon reached the conviction that much in the stories of the bible could not be true. the consequence was a positively fanatic freethinking coupled with the impression that youth is intentionally being deceived by the state through lies: it was a crushing impression. [2]

i remember that feeling. by 15 i was convinced the world was corrupt from end to end. that's why movies like *the matrix* have such resonance. every kid grows up in a fake world. in a way it would be easier if the forces behind it were as clearly differentiated as a bunch of evil machines, and one could make a clean break just by taking a pill.

protection

if you ask adults why they lie to kids, the most common reason they give is to protect them. and kids do need protecting. the environment you want to create for a newborn child will be quite unlike the streets of a big city.

that seems so obvious it seems wrong to call it a lie. it's certainly not a bad lie to tell, to give a baby the impression the world is quiet and warm and safe. but this harmless type of lie can turn sour if left unexamined.

imagine if you tried to keep someone in as protected an environment as a newborn till age 18. to mislead someone so grossly about the world would seem not protection but abuse. that's an extreme example, of course; when parents do that sort of thing it becomes national news. but you see the same problem on a smaller scale in the malaise teenagers feel in suburbia.

the main purpose of suburbia is to provide a protected environment for children to grow up in. and it seems great for 10 year olds. i liked living in suburbia when i was 10. i didn't notice how sterile it was.

mothers. so what's going on? there does seem to be a universal taboo against sex with prepubescent children. one can imagine evolutionary reasons for that. and i think this is the main reason parents in industrialized societies dislike teenage kids having sex. they still think of them as children, even though biologically they're not, so the taboo against child sex still has force.

one thing adults conceal about sex they also conceal about drugs: that it can cause great pleasure. that's what makes sex and drugs so dangerous. the desire for them can cloud one's judgement—which is especially frightening when the judgement being clouded is the already wretched judgement of a teenage kid.

here parents' desires conflict. older societies told kids they had bad judgement, but modern parents want their children to be confident. this may well be a better plan than the old one of putting them in their place, but it has the side effect that after having implicitly lied to kids about how good their judgement is, we then have to lie again about all the things they might get into trouble with if they believed us.

if parents told their kids the truth about sex and drugs, it would be: the reason you should avoid these things is that you have lousy judgement. people with twice your experience still get burned by them. but this may be one of those cases where the truth wouldn't be convincing, because one of the symptoms of bad judgement is believing you have good judgement. when you're too weak to lift something, you can tell, but when you're making a decision impetuously, you're all the more sure of it.

innocence

another reason parents don't want their kids having sex is that they want to keep them innocent. adults have a certain model of how kids are supposed to behave, and it's different from what they expect of other adults.

one of the most obvious differences is the words kids are allowed to use. most parents use words when talking to other adults that they wouldn't want their kids using. they try to hide even the existence of these words for as long as they can. and this is another of those conspiracies everyone participates in: everyone knows you're not supposed to swear in front of kids.

i've never heard more different explanations for anything parents tell kids than why they shouldn't swear. every parent i know forbids their children to swear, and yet no two of them have the same justification. it's clear most start with not wanting kids to swear, then make up the reason afterward.

so my theory about what's going on is that the *function* of swearwords is to mark the speaker as an adult. there's no difference in the meaning of "shit" and "poopoo." so why should one be ok for kids to say and one forbidden? the only explanation is: by definition. [3]

why does it bother adults so much when kids do things reserved for adults? the idea of a foul-mouthed, cynical 10 year old leaning against a lamppost with a cigarette hanging out of the corner of his mouth is very disconcerting. but why?

one reason we want kids to be innocent is that we're programmed to like certain kinds of helplessness. i've several times heard mothers say they deliberately refrained from correcting their young children's mispronunciations because they were so cute. and if you think about it, cuteness is helplessness. toys and cartoon characters meant to be cute always have clueless expressions and stubby, ineffectual limbs.

it's not surprising we'd have an inborn desire to love and protect helpless creatures, considering human offspring are so helpless for so long. without the helplessness that makes kids cute, they'd be very an-

think for themselves. but if you tell a kid they're a member of a certain group, that seems nearly impossible to shake.

this despite the fact that it can be one of the most premeditated lies parents tell. when parents are of different religions, they'll often agree between themselves that their children will be "raised as xes." and it works. the kids obligingly grow up considering themselves as xes, despite the fact that if their parents had chosen the other way, they'd have grown up considering themselves as ys.

one reason this works so well is the second kind of lie involved. the truth is common property. you can't distinguish your group by doing things that are rational, and believing things that are true. if you want to set yourself apart from other people, you have to do things that are arbitrary, and believe things that are false. and after having spent their whole lives doing things that are arbitrary and believing things that are false, and being regarded as odd by "outsiders" on that account, the cognitive dissonance pushing children to regard themselves as xes must be enormous. if they aren't an x, why are they attached to all these arbitrary beliefs and customs? if they aren't an x, why do all the non-xes call them one?

this form of lie is not without its uses. you can use it to carry a payload of beneficial beliefs, and they will also become part of the child's identity. you can tell the child that in addition to never wearing the color yellow, believing the world was created by a giant rabbit, and always snapping their fingers before eating fish, xes are also particularly honest and industrious. then x children will grow up feeling it's part of their identity to be honest and industrious.

this probably accounts for a lot of the spread of modern religions, and explains why their doctrines are a combination of the useful and the bizarre. the bizarre half is what makes the religion stick, and the useful half is the payload. [6]

authority

one of the least excusable reasons adults lie to kids is to maintain power over them. sometimes these lies are truly sinister, like a child molester telling his victims they'll get in trouble if they tell anyone what happened to them. others seem more innocent; it depends how badly adults lie to maintain their power, and what they use it for.

most adults make some effort to conceal their flaws from children. usually their motives are mixed. for example, a father who has an affair generally conceals it from his children. his motive is partly that it would worry them, partly that this would introduce the topic of sex, and partly (a larger part than he would admit) that he doesn't want to tarnish himself in their eyes.

if you want to learn what lies are told to kids, read almost any book written to teach them about "issues." [7] peter mayle wrote one called *why are we getting a divorce?* it begins with the three most important things to remember about divorce, one of which is:

> you shouldn't put the blame on one parent, because divorce is never only one person's fault. [8]

really? when a man runs off with his secretary, is it always partly his wife's fault? but i can see why mayle might have said this. maybe it's more important for kids to respect their parents than to know the truth about them.

but because adults conceal their flaws, and at the same time insist on high standards of behavior for kids, a lot of kids grow up feeling they fall hopelessly short. they walk around feeling horribly evil for having used a swearword, while in fact most of the adults around them are doing much worse things.

peace

of all the reasons we lie to kids, the most powerful is probably the same mundane reason they lie to us.

often when we lie to people it's not part of any conscious strategy, but because they'd react violently to the truth. kids, almost by definition, lack self-control. they react violently to things—and so they get lied to a lot. [9]

a few thanksgivings ago, a friend of mine found himself in a situation that perfectly illustrates the complex motives we have when we lie to kids. as the roast turkey appeared on the table, his alarmingly perceptive 5 year old son suddenly asked if the turkey had wanted to die. foreseeing disaster, my friend and his wife rapidly improvised: yes, the turkey had wanted to die, and in fact had lived its whole life with the aim of being their thanksgiving dinner. and that (phew) was the end of that.

whenever we lie to kids to protect them, we're usually also lying to keep the peace.

one consequence of this sort of calming lie is that we grow up thinking horrible things are normal. it's hard for us to feel a sense of urgency as adults over something we've literally been trained not to worry about. when i was about 10 i saw a documentary on pollution that put me into a panic. it seemed the planet was being irretrievably ruined. i went to my mother afterward to ask if this was so. i don't remember what she said, but she made me feel better, so i stopped worrying about it.

that was probably the best way to handle a frightened 10 year old. but we should understand the price. this sort of lie is one of the main reasons bad things persist: we're all trained to ignore them.

detox

a sprinter in a race almost immediately enters a state called "oxygen debt." his body switches to an emergency source of energy that's faster than regular aerobic respiration. but this process builds up waste products that ultimately require extra oxygen to break down, so at the end of the race he has to stop and pant for a while to recover.

we arrive at adulthood with a kind of truth debt. we were told a lot of lies to get us (and our parents) through our childhood. some may have been necessary. some probably weren't. but we all arrive at adulthood with heads full of lies.

there's never a point where the adults sit you down and explain all the lies they told you. they've forgotten most of them. so if you're going to clear these lies out of your head, you're going to have to do it yourself.

few do. most people go through life with bits of packing material adhering to their minds and never know it. you probably never can completely undo the effects of lies you were told as a kid, but it's worth trying. i've found that whenever i've been able to undo a lie i was told, a lot of other things fell into place.

fortunately, once you arrive at adulthood you get a valuable new resource you can use to figure out what lies you were told. you're now one of the liars. you get to watch behind the scenes as adults spin the world for the next generation of kids.

the first step in clearing your head is to realize how far you are from a neutral observer. when i left high school i was, i thought, a complete skeptic. i'd realized high school was crap. i thought i was ready to question everything i knew. but among the many other things i was ignorant of was how much debris there already was in my head. it's not enough to consider your mind a blank slate. you have to consciously erase it.

(i wrote this talk for a high school. i never actually gave it, because the school authorities vetoed the plan to invite me.)

when i said i was speaking at a high school, my friends were curious. what will you say to high school students? so i asked them, what do you wish someone had told you in high school? their answers were remarkably similar. so i'm going to tell you what we all wish someone had told us.

i'll start by telling you something you don't have to know in high school: what you want to do with your life. people are always asking you this, so you think you're supposed to have an answer. but adults ask this mainly as a conversation starter. they want to know what sort of person you are, and this question is just to get you talking. they ask it the way you might poke a hermit crab in a tide pool, to see what it does.

if i were back in high school and someone asked about my plans, i'd say that my first priority was to learn what the options were. you don't need to be in a rush to choose your life's work. what you need to do is discover what you like. you have to work on stuff you like if you want to be good at what you do.

it might seem that nothing would be easier than deciding what you like, but it turns out to be hard, partly because it's hard to get an accurate picture of most jobs. being a doctor is not the way it's portrayed on tv. fortunately you can also watch real doctors, by volunteering in hospitals.[1]

but there are other jobs you can't learn about, because no one is doing them yet. most of the work i've done in the last ten years didn't exist when i was in high school. the world changes fast, and the rate at which it changes is itself speeding up. in such a world it's not a good idea to have fixed plans.

and yet every may, speakers all over the country fire up the standard graduation speech, the theme of which is: don't give up on your dreams. i know what they mean, but this is a bad way to put it, because it implies you're supposed to be bound by some plan you made early on. the computer world has a name for this: premature optimization. and it is synonymous with disaster. these speakers would do better to say simply, don't give up.

what they really mean is, don't get demoralized. don't think that you can't do what other people can. and i agree you shouldn't underestimate your potential. people who've done great things tend to seem as if they were a race apart. and most biographies only exaggerate this illusion, partly due to the worshipful attitude biographers inevitably sink into, and partly because, knowing how the story ends, they can't help streamlining the plot till it seems like the subject's life was a matter of destiny, the mere unfolding of some innate genius. in fact i suspect if you had the sixteen year old shakespeare or einstein in school with you, they'd seem impressive, but not totally unlike your other friends.

which is an uncomfortable thought. if they were just like us, then they had to work very hard to do what they did. and that's one reason we like to believe in genius. it gives us an excuse for being lazy. if these guys were able to do what they did only because of some magic shakespeareness or einsteinness, then it's not our fault if we can't do something as good.

i'm not saying there's no such thing as genius. but if you're trying to choose between two theories and one gives you an excuse for being lazy, the other one is probably right.

so far we've cut the standard graduation speech down from "don't give up on your dreams" to "what someone else can do, you can do." but it needs to be cut still further. there is *some* variation in natural ability. most people overestimate its role, but it does exist. if i were talking to a guy four feet tall whose ambition was to play in the nba, i'd feel pretty stupid saying, you can do anything if you really try.[2]

i'm not saying this is the only way to be happy. just that some kinds of worry are not as bad as they sound.

ambition

in practice, "stay upwind" reduces to "work on hard problems." and you can start today. i wish i'd grasped that in high school.

most people like to be good at what they do. in the so-called real world this need is a powerful force. but high school students rarely benefit from it, because they're given a fake thing to do. when i was in high school, i let myself believe that my job was to be a high school student. and so i let my need to be good at what i did be satisfied by merely doing well in school.

if you'd asked me in high school what the difference was between high school kids and adults, i'd have said it was that adults had to earn a living. wrong. it's that adults take responsibility for themselves. making a living is only a small part of it. far more important is to take intellectual responsibility for oneself.

if i had to go through high school again, i'd treat it like a day job. i don't mean that i'd slack in school. working at something as a day job doesn't mean doing it badly. it means not being defined by it. i mean i wouldn't think of myself as a high school student, just as a musician with a day job as a waiter doesn't think of himself as a waiter.[3] and when i wasn't working at my day job i'd start trying to do real work.

when i ask people what they regret most about high school, they nearly all say the same thing: that they wasted so much time. if you're wondering what you're doing now that you'll regret most later, that's probably it.[4]

some people say this is inevitable — that high school students aren't capable of getting anything done yet. but i don't think this is true. and the proof is that you're bored. you probably weren't bored when you were eight. when you're eight it's called "playing" instead of "hanging out," but it's the same thing. and when i was eight, i was rarely bored. give me a back yard and a few other kids and i could play all day.

the reason this got stale in middle school and high school, i now realize, is that i was ready for something else. childhood was getting old.

i'm not saying you shouldn't hang out with your friends — that you should all become humorless little robots who do nothing but work. hanging out with friends is like chocolate cake. you enjoy it more if you eat it occasionally than if you eat nothing but chocolate cake for every meal. no matter how much you like chocolate cake, you'll be pretty queasy after the third meal of it. and that's what the malaise one feels in high school is: mental queasiness.[5]

you may be thinking, we have to do more than get good grades. we have to have *extracurricular activities*. but you know perfectly well how bogus most of these are. collecting donations for a charity is an admirable thing to do, but it's not *hard*. it's not getting something done. what i mean by getting something done is learning how to write well, or how to program computers, or what life was really like in preindustrial societies, or how to draw the human face from life. this sort of thing rarely translates into a line item on a college application.

corruption

it's dangerous to design your life around getting into college, because the people you have to impress to get into college are not a very discerning audience. at most colleges, it's not the professors who dec-

curiosity turns work into play. for einstein, relativity wasn't a book full of hard stuff he had to learn for an exam. it was a mystery he was trying to solve. so it probably felt like less work to him to invent it than it would seem to someone now to learn it in a class.

one of the most dangerous illusions you get from school is the idea that doing great things requires a lot of discipline. most subjects are taught in such a boring way that it's only by discipline that you can flog yourself through them. so i was surprised when, early in college, i read a quote by wittgenstein saying that he had no self-discipline and had never been able to deny himself anything, not even a cup of coffee.

now i know a number of people who do great work, and it's the same with all of them. they have little discipline. they're all terrible procrastinators and find it almost impossible to make themselves do anything they're not interested in. one still hasn't sent out his half of the thank-you notes from his wedding, four years ago. another has 26,000 emails in her inbox.

i'm not saying you can get away with zero self-discipline. you probably need about the amount you need to go running. i'm often reluctant to go running, but once i do, i enjoy it. and if i don't run for several days, i feel ill. it's the same with people who do great things. they know they'll feel bad if they don't work, and they have enough discipline to get themselves to their desks to start working. but once they get started, interest takes over, and discipline is no longer necessary.

do you think shakespeare was gritting his teeth and diligently trying to write great literature? of course not. he was having fun. that's why he's so good.

if you want to do good work, what you need is a great curiosity about a promising question. the critical moment for einstein was when he looked at maxwell's equations and said, what the hell is going on here?

it can take years to zero in on a productive question, because it can take years to figure out what a subject is really about. to take an extreme example, consider math. most people think they hate math, but the boring stuff you do in school under the name "mathematics" is not at all like what mathematicians do.

the great mathematician g. h. hardy said he didn't like math in high school either. he only took it up because he was better at it than the other students. only later did he realize math was interesting — only later did he start to ask questions instead of merely answering them correctly.

when a friend of mine used to grumble because he had to write a paper for school, his mother would tell him: find a way to make it interesting. that's what you need to do: find a question that makes the world interesting. people who do great things look at the same world everyone else does, but notice some odd detail that's compellingly mysterious.

and not only in intellectual matters. henry ford's great question was, why do cars have to be a luxury item? what would happen if you treated them as a commodity? franz beckenbauer's was, in effect, why does everyone have to stay in his position? why can't defenders score goals too?

now

if it takes years to articulate great questions, what do you do now, at sixteen? work toward finding one. great questions don't appear suddenly. they gradually congeal in your head. and what makes them congeal is experience. so the way to find great questions is not to search for them — not to wander about thinking, what great discovery shall i make? you can't answer that; if you could, you'd have made it.

this may sound like bullshit. i'm just a minor, you may think, i have no money, i have to live at home, i have to do what adults tell me all day long. well, most adults labor under restrictions just as cumbersome, and they manage to get things done. if you think it's restrictive being a kid, imagine having kids.

the only real difference between adults and high school kids is that adults realize they need to get things done, and high school kids don't. that realization hits most people around 23. but i'm letting you in on the secret early. so get to work. maybe you can be the first generation whose greatest regret from high school isn't how much time you wasted.

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notes

1 a doctor friend warns that even this can give an inaccurate picture. "who knew how much time it would take up, how little autonomy one would have for endless years of training, and how unbelievably annoying it is to carry a beeper?"

2 his best bet would probably be to become dictator and intimidate the nba into letting him play. so far the closest anyone has come is secretary of labor.

3 a day job is one you take to pay the bills so you can do what you really want, like play in a band, or invent relativity.

treating high school as a day job might actually make it easier for some students to get good grades. if you treat your classes as a game, you won't be demoralized if they seem pointless.

however bad your classes, you need to get good grades in them to get into a decent college. and that *is* worth doing, because universities are where a lot of the clumps of smart people are these days.

4 the second biggest regret was caring so much about unimportant things. and especially about what other people thought of them.

i think what they really mean, in the latter case, is caring what random people thought of them. adults care just as much what other people think, but they get to be more selective about the other people.

i have about thirty friends whose opinions i care about, and the opinion of the rest of the world barely affects me. the problem in high school is that your peers are chosen for you by accidents of age and geography, rather than by you based on respect for their judgement.

5 the key to wasting time is distraction. without distractions it's too obvious to your brain that you're not doing anything with it, and you start to feel uncomfortable. if you want to measure how dependent you've become on distractions, try this experiment: set aside a chunk of time on a weekend and sit alone and think. you can have a notebook to write your thoughts down in, but nothing else: no friends, tv, music, phone, im, email, web, games, books, newspapers, or magazines. within an hour most people will feel a strong craving for distraction.

6 i don't mean to imply that the only function of prep schools is to trick admissions officers. they also generally provide a better education. but try this thought experiment: suppose prep schools supplied the same superior education but had a tiny (.001) negative effect on college admissions. how many parents would still send their kids to them?

it might also be argued that kids who went to prep schools, because they've learned more, *are* better college candidates. but this seems empirically false. what you learn in even the best high school is rounding error compared to what you learn in college. public school kids arrive at college with a slight disadvantage, but they start to pull ahead in the sophomore year.

one of the biggest things holding people back from doing great work is the fear of making something lame. and this fear is not an irrational one. many great projects go through a stage early on where they don't seem very impressive, even to their creators. you have to push through this stage to reach the great work that lies beyond. but many people don't. most people don't even reach the stage of making something they're embarrassed by, let alone continue past it. they're too frightened even to start.

imagine if we could turn off the fear of making something lame. imagine how much more we'd do.

is there any hope of turning it off? i think so. i think the habits at work here are not very deeply rooted.

making new things is itself a new thing for us as a species. it has always happened, but till the last few centuries it happened so slowly as to be invisible to individual humans. and since we didn't need customs for dealing with new ideas, we didn't develop any.

we just don't have enough experience with early versions of ambitious projects to know how to respond to them. we judge them as we would judge more finished work, or less ambitious projects. we don't realize they're a special case.

or at least, most of us don't. one reason i'm confident we can do better is that it's already starting to happen. there are already a few places that are living in the future in this respect. silicon valley is one of them: an unknown person working on a strange-sounding idea won't automatically be dismissed the way they would back home. in silicon valley, people have learned how dangerous that is.

the right way to deal with new ideas is to treat them as a challenge to your imagination — not just to have lower standards, but to switch polarity entirely, from listing the reasons an idea won't work to trying to think of ways it could. that's what i do when i meet people with new ideas. i've become quite good at it, but i've had a lot of practice. being a partner at y combinator means being practically immersed in strange-sounding ideas proposed by unknown people. every six months you get thousands of new ones thrown at you and have to sort through them, knowing that in a world with a power-law distribution of outcomes, it will be painfully obvious if you miss the needle in this haystack. optimism becomes urgent.

but i'm hopeful that, with time, this kind of optimism can become widespread enough that it becomes a social custom, not just a trick used by a few specialists. it is after all an extremely lucrative trick, and those tend to spread quickly.

of course, inexperience is not the only reason people are too harsh on early versions of ambitious projects. they also do it to seem clever. and in a field where the new ideas are risky, like startups, those who dismiss them are in fact more likely to be right. just not when their predictions are weighted by outcome.

but there is another more sinister reason people dismiss new ideas. if you try something ambitious, many of those around you will hope, consciously or unconsciously, that you'll fail. they worry that if you try something ambitious and succeed, it will put you above them. in some countries this is not just an individual failing but part of the national culture.

i wouldn't claim that people in silicon valley overcome these impulses because they're morally better. [1] the reason many hope you'll succeed is that they hope to rise with you. for investors this incentive is particularly explicit. they want you to succeed because they hope you'll make them rich in the process. but many other people you meet can hope to benefit in some way from your success. at the very least they'll be able to say, when you're famous, that they've known you since way back.

teachers are in effect a special case of colleagues. it's a teacher's job both to see the promise of early work and to encourage you to continue. but teachers who are good at this are unfortunately quite rare, so if you have the opportunity to learn from one, take it. [3]

for some it might work to rely on sheer discipline: to tell yourself that you just have to press on through the initial crap phase and not get discouraged. but like a lot of "just tell yourself" advice, this is harder than it sounds. and it gets still harder as you get older, because your standards rise. the old do have one compensating advantage though: they've been through this before.

it can help if you focus less on where you are and more on the rate of change. you won't worry so much about doing bad work if you can see it improving. obviously the faster it improves, the easier this is. so when you start something new, it's good if you can spend a lot of time on it. that's another advantage of being young: you tend to have bigger blocks of time.

another common trick is to start by considering new work to be of a different, less exacting type. to start a painting saying that it's just a sketch, or a new piece of software saying that it's just a quick hack. then you judge your initial results by a lower standard. once the project is rolling you can sneakily convert it to something more. [4]

this will be easier if you use a medium that lets you work fast and doesn't require too much commitment up front. it's easier to convince yourself that something is just a sketch when you're drawing in a notebook than when you're carving stone. plus you get initial results faster. [5] [6]

it will be easier to try out a risky project if you think of it as a way to learn and not just as a way to make something. then even if the project truly is a failure, you'll still have gained by it. if the problem is sharply enough defined, failure itself is knowledge: if the theorem you're trying to prove turns out to be false, or you use a structural member of a certain size and it fails under stress, you've learned something, even if it isn't what you wanted to learn. [7]

one motivation that works particularly well for me is curiosity. i like to try new things just to see how they'll turn out. we started y combinator in this spirit, and it was one of main things that kept me going while i was working on bel. having worked for so long with various dialects of lisp, i was very curious to see what its inherent shape was: what you'd end up with if you followed the axiomatic approach all the way.

but it's a bit strange that you have to play mind games with yourself to avoid being discouraged by lame-looking early efforts. the thing you're trying to trick yourself into believing is in fact the truth. a lame-looking early version of an ambitious project truly is more valuable than it seems. so the ultimate solution may be to teach yourself that.

one way to do it is to study the histories of people who've done great work. what were they thinking early on? what was the very first thing they did? it can sometimes be hard to get an accurate answer to this question, because people are often embarrassed by their earliest work and make little effort to publish it. (they too misjudge it.) but when you can get an accurate picture of the first steps someone made on the path to some great work, they're often pretty feeble. [8]

perhaps if you study enough such cases, you can teach yourself to be a better judge of early work. then you'll be immune both to other people's skepticism and your own fear of making something lame. you'll see early work for what it is.

curiously enough, the solution to the problem of judging early work too harshly is to realize that our attitudes toward it are themselves early work. holding everything to the same standard is a crude vers-

when i was young, i thought old people had everything figured out. now that i'm old, i know this isn't true.

i constantly feel like a noob. it seems like i'm always talking to some startup working in a new field i know nothing about, or reading a book about a topic i don't understand well enough, or visiting some new country where i don't know how things work.

it's not pleasant to feel like a noob. and the word "noob" is certainly not a compliment. and yet today i realized something encouraging about being a noob: the more of a noob you are locally, the less of a noob you are globally.

for example, if you stay in your home country, you'll feel less of a noob than if you move to farawavia, where everything works differently. and yet you'll know more if you move. so the feeling of being a noob is inversely correlated with actual ignorance.

but if the feeling of being a noob is good for us, why do we dislike it? what evolutionary purpose could such an aversion serve?

i think the answer is that there are two sources of feeling like a noob: being stupid, and doing something novel. our dislike of feeling like a noob is our brain telling us "come on, come on, figure this out." which was the right thing to be thinking for most of human history. the life of hunter-gatherers was complex, but it didn't change as much as life does now. they didn't suddenly have to figure out what to do about cryptocurrency. so it made sense to be biased toward competence at existing problems over the discovery of new ones. it made sense for humans to dislike the feeling of being a noob, just as, in a world where food was scarce, it made sense for them to dislike the feeling of being hungry.

now that too much food is more of a problem than too little, our dislike of feeling hungry leads us astray. and i think our dislike of feeling like a noob does too.

though it feels unpleasant, and people will sometimes ridicule you for it, the more you feel like a noob, the better.

in the science fiction books i read as a kid, reading had often been replaced by some more efficient way of acquiring knowledge. mysterious "tapes" would load it into one's brain like a program being loaded into a computer.

that sort of thing is unlikely to happen anytime soon. not just because it would be hard to build a replacement for reading, but because even if one existed, it would be insufficient. reading about x doesn't just teach you about x; it also teaches you how to write. [1]

would that matter? if we replaced reading, would anyone need to be good at writing?

the reason it would matter is that writing is not just a way to convey ideas, but also a way to have them.

a good writer doesn't just think, and then write down what he thought, as a sort of transcript. a good writer will almost always discover new things in the process of writing. and there is, as far as i know, no substitute for this kind of discovery. talking about your ideas with other people is a good way to develop them. but even after doing this, you'll find you still discover new things when you sit down to write. there is a kind of thinking that can only be done by writing.

there are of course kinds of thinking that can be done without writing. if you don't need to go too deeply into a problem, you can solve it without writing. if you're thinking about how two pieces of machinery should fit together, writing about it probably won't help much. and when a problem can be described formally, you can sometimes solve it in your head. but if you need to solve a complicated, ill-defined problem, it will almost always help to write about it. which in turn means that someone who's not good at writing will almost always be at a disadvantage in solving such problems.

you can't think well without writing well, and you can't write well without reading well. and i mean that last "well" in both senses. you have to be good at reading, and read good things. [2]

people who just want information may find other ways to get it. but people who want to have ideas can't afford to.

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notes

1 audiobooks can give you examples of good writing, but having them read to you doesn't teach you as much about writing as reading them yourself.

2 by "good at reading" i don't mean good at the mechanics of reading. you don't have to be good at extracting words from the page so much as extracting meaning from the words.

if you asked people what was special about einstein, most would say that he was really smart. even the ones who tried to give you a more sophisticated-sounding answer would probably think this first. till a few years ago i would have given the same answer myself. but that wasn't what was special about einstein. what was special about him was that he had important new ideas. being very smart was a necessary precondition for having those ideas, but the two are not identical.

it may seem a hair-splitting distinction to point out that intelligence and its consequences are not identical, but it isn't. there's a big gap between them. anyone who's spent time around universities and research labs knows how big. there are a lot of genuinely smart people who don't achieve very much.

i grew up thinking that being smart was the thing most to be desired. perhaps you did too. but i bet it's not what you really want. imagine you had a choice between being really smart but discovering nothing new, and being less smart but discovering lots of new ideas. surely you'd take the latter. i would. the choice makes me uncomfortable, but when you see the two options laid out explicitly like that, it's obvious which is better.

the reason the choice makes me uncomfortable is that being smart still feels like the thing that matters, even though i know intellectually that it isn't. i spent so many years thinking it was. the circumstances of childhood are a perfect storm for fostering this illusion. intelligence is much easier to measure than the value of new ideas, and you're constantly being judged by it. whereas even the kids who will ultimately discover new things aren't usually discovering them yet. for kids that way inclined, intelligence is the only game in town.

there are more subtle reasons too, which persist long into adulthood. intelligence wins in conversation, and thus becomes the basis of the dominance hierarchy. [1] plus having new ideas is such a new thing historically, and even now done by so few people, that society hasn't yet assimilated the fact that this is the actual destination, and intelligence merely a means to an end. [2]

why do so many smart people fail to discover anything new? viewed from that direction, the question seems a rather depressing one. but there's another way to look at it that's not just more optimistic, but more interesting as well. clearly intelligence is not the only ingredient in having new ideas. what are the other ingredients? are they things we could cultivate?

because the trouble with intelligence, they say, is that it's mostly inborn. the evidence for this seems fairly convincing, especially considering that most of us don't want it to be true, and the evidence thus has to face a stiff headwind. but i'm not going to get into that question here, because it's the other ingredients in new ideas that i care about, and it's clear that many of them can be cultivated.

that means the truth is excitingly different from the story i got as a kid. if intelligence is what matters, and also mostly inborn, the natural consequence is a sort of *brave new world* fatalism. the best you can do is figure out what sort of work you have an "aptitude" for, so that whatever intelligence you were born with will at least be put to the best use, and then work as hard as you can at it. whereas if intelligence isn't what matters, but only one of several ingredients in what does, and many of those aren't inborn, things get more interesting. you have a lot more control, but the problem of how to arrange your life becomes that much more complicated.

so what are the other ingredients in having new ideas? the fact that i can even ask this question proves the point i raised earlier — that society hasn't assimilated the fact that it's this and not intelligence that matters. otherwise we'd all know the answers to such a fundamental question. [3]

i'm not going to try to provide a complete catalogue of the other ingredients here. this is the first time i've posed the question to myself this way, and i think it may take a while to answer. but i wrote

about: it's neither separable from intelligence, nor responsible for all the difference between intelligence and having new ideas.

4 curiously enough, this essay is an example. it started out as an essay about writing ability. but when i came to the distinction between intelligence and having new ideas, that seemed so much more important that i turned the original essay inside out, making that the topic and my original topic one of the points in it. as in many other fields, that level of reworking is easier to contemplate once you've had a lot of practice.

thanks to trevor blackwell, patrick collison, jessica livingston, robert morris, michael nielsen, and lisa randall for reading drafts of this.

when i was in high school i spent a lot of time imitating bad writers. what we studied in english classes was mostly fiction, so i assumed that was the highest form of writing. mistake number one. the stories that seemed to be most admired were ones in which people suffered in complicated ways. anything funny or gripping was ipso facto suspect, unless it was old enough to be hard to understand, like shakespeare or chaucer. mistake number two. the ideal medium seemed the short story, which i've since learned had quite a brief life, roughly coincident with the peak of magazine publishing. but since their size made them perfect for use in high school classes, we read a lot of them, which gave us the impression the short story was flourishing. mistake number three. and because they were so short, nothing really had to happen; you could just show a randomly truncated slice of life, and that was considered advanced. mistake number four. the result was that i wrote a lot of stories in which nothing happened except that someone was unhappy in a way that seemed deep.

for most of college i was a philosophy major. i was very impressed by the papers published in philosophy journals. they were so beautifully typeset, and their tone was just captivating—alternately casual and buffer-overflowingly technical. a fellow would be walking along a street and suddenly modality qua modality would spring upon him. i didn't ever quite understand these papers, but i figured i'd get around to that later, when i had time to reread them more closely. in the meantime i tried my best to imitate them. this was, i can now see, a doomed undertaking, because they weren't really saying anything. no philosopher ever refuted another, for example, because no one said anything definite enough to refute. needless to say, my imitations didn't say anything either.

in grad school i was still wasting time imitating the wrong things. there was then a fashionable type of program called an expert system, at the core of which was something called an inference engine. i looked at what these things did and thought "i could write that in a thousand lines of code." and yet eminent professors were writing books about them, and startups were selling them for a year's salary a copy. what an opportunity, i thought; these impressive things seem easy to me; i must be pretty sharp. wrong. it was simply a fad. the books the professors wrote about expert systems are now ignored. they were not even on a *path* to anything interesting. and the customers paying so much for them were largely the same government agencies that paid thousands for screwdrivers and toilet seats.

how do you avoid copying the wrong things? copy only what you genuinely like. that would have saved me in all three cases. i didn't enjoy the short stories we had to read in english classes; i didn't learn anything from philosophy papers; i didn't use expert systems myself. i believed these things were good because they were admired.

it can be hard to separate the things you like from the things you're impressed with. one trick is to ignore presentation. whenever i see a painting impressively hung in a museum, i ask myself: how much would i pay for this if i found it at a garage sale, dirty and frameless, and with no idea who painted it? if you walk around a museum trying this experiment, you'll find you get some truly startling results. don't ignore this data point just because it's an outlier.

another way to figure out what you like is to look at what you enjoy as guilty pleasures. many things people like, especially if they're young and ambitious, they like largely for the feeling of virtue in liking them. 99% of people reading *ulysses* are thinking "i'm reading *ulysses*" as they do it. a guilty pleasure is at least a pure one. what do you read when you don't feel up to being virtuous? what kind of book do you read and feel sad that there's only half of it left, instead of being impressed that you're half way through? that's what you really like.

even when you find genuinely good things to copy, there's another pitfall to be avoided. be careful to copy what makes them good, rather than their flaws. it's easy to be drawn into imitating flaws, because

i've discovered a handy test for figuring out what you're addicted to. imagine you were going to spend the weekend at a friend's house on a little island off the coast of maine. there are no shops on the island and you won't be able to leave while you're there. also, you've never been to this house before, so you can't assume it will have more than any house might.

what, besides clothes and toiletries, do you make a point of packing? that's what you're addicted to. for example, if you find yourself packing a bottle of vodka (just in case), you may want to stop and think about that.

for me the list is four things: books, earplugs, a notebook, and a pen.

there are other things i might bring if i thought of it, like music, or tea, but i can live without them. i'm not so addicted to caffeine that i wouldn't risk the house not having any tea, just for a weekend.

quiet is another matter. i realize it seems a bit eccentric to take earplugs on a trip to an island off the coast of maine. if anywhere should be quiet, that should. but what if the person in the next room snored? what if there was a kid playing basketball? (thump, thump, thump... thump.) why risk it? earplugs are small.

sometimes i can think with noise. if i already have momentum on some project, i can work in noisy places. i can edit an essay or debug code in an airport. but airports are not so bad: most of the noise is whitish. i couldn't work with the sound of a sitcom coming through the wall, or a car in the street playing thump-thump music.

and of course there's another kind of thinking, when you're starting something new, that requires complete quiet. you never know when this will strike. it's just as well to carry plugs.

the notebook and pen are professional equipment, as it were. though actually there is something druglike about them, in the sense that their main purpose is to make me feel better. i hardly ever go back and read stuff i write down in notebooks. it's just that if i can't write things down, worrying about remembering one idea gets in the way of having the next. pen and paper wick ideas.

the best notebooks i've found are made by a company called miquelrius. i use their smallest size, which is about 2.5 x 4 in. the secret to writing on such narrow pages is to break words only when you run out of space, like a latin inscription. i use the cheapest plastic bic ballpoints, partly because their gluey ink doesn't seep through pages, and partly so i don't worry about losing them.

i only started carrying a notebook about three years ago. before that i used whatever scraps of paper i could find. but the problem with scraps of paper is that they're not ordered. in a notebook you can guess what a scribble means by looking at the pages around it. in the scrap era i was constantly finding notes i'd written years before that might say something i needed to remember, if i could only figure out what.

as for books, i know the house would probably have something to read. on the average trip i bring four books and only read one of them, because i find new books to read en route. really bringing books is insurance.

i realize this dependence on books is not entirely good—that what i need them for is distraction. the books i bring on trips are often quite virtuous, the sort of stuff that might be assigned reading in a college class. but i know my motives aren't virtuous. i bring books because if the world gets boring i need to be able to slip into another distilled by some writer. it's like eating jam when you know you should be eating fruit.

july 2010

i realized recently that what one thinks about in the shower in the morning is more important than i'd thought. i knew it was a good time to have ideas. now i'd go further: now i'd say it's hard to do a really good job on anything you don't think about in the shower.

everyone who's worked on difficult problems is probably familiar with the phenomenon of working hard to figure something out, failing, and then suddenly seeing the answer a bit later while doing something else. there's a kind of thinking you do without trying to. i'm increasingly convinced this type of thinking is not merely helpful in solving hard problems, but necessary. the tricky part is, you can only control it indirectly. [1]

i think most people have one top idea in their mind at any given time. that's the idea their thoughts will drift toward when they're allowed to drift freely. and this idea will thus tend to get all the benefit of that type of thinking, while others are starved of it. which means it's a disaster to let the wrong idea become the top one in your mind.

what made this clear to me was having an idea i didn't want as the top one in my mind for two long stretches.

i'd noticed startups got way less done when they started raising money, but it was not till we ourselves raised money that i understood why. the problem is not the actual time it takes to meet with investors. the problem is that once you start raising money, raising money becomes the top idea in your mind. that becomes what you think about when you take a shower in the morning. and that means other questions aren't.

i'd hated raising money when i was running viaweb, but i'd forgotten why i hated it so much. when we raised money for y combinator, i remembered. money matters are particularly likely to become the top idea in your mind. the reason is that they have to be. it's hard to get money. it's not the sort of thing that happens by default. it's not going to happen unless you let it become the thing you think about in the shower. and then you'll make little progress on anything else you'd rather be working on. [2]

(i hear similar complaints from friends who are professors. professors nowadays seem to have become professional fundraisers who do a little research on the side. it may be time to fix that.)

the reason this struck me so forcibly is that for most of the preceding 10 years i'd been able to think about what i wanted. so the contrast when i couldn't was sharp. but i don't think this problem is unique to me, because just about every startup i've seen grinds to a halt when they start raising money — or talking to acquirers.

you can't directly control where your thoughts drift. if you're controlling them, they're not drifting. but you can control them indirectly, by controlling what situations you let yourself get into. that has been the lesson for me: be careful what you let become critical to you. try to get yourself into situations where the most urgent problems are ones you want to think about.

you don't have complete control, of course. an emergency could push other thoughts out of your head. but barring emergencies you have a good deal of indirect control over what becomes the top idea in your mind.

i've found there are two types of thoughts especially worth avoiding — thoughts like the nile perch in the way they push out more interesting ideas. one i've already mentioned: thoughts about money. gett-

the most impressive people i know are all terrible procrastinators. so could it be that procrastination isn't always bad?

most people who write about procrastination write about how to cure it. but this is, strictly speaking, impossible. there are an infinite number of things you could be doing. no matter what you work on, you're not working on everything else. so the question is not how to avoid procrastination, but how to procrastinate well.

there are three variants of procrastination, depending on what you do instead of working on something: you could work on (a) nothing, (b) something less important, or (c) something more important. that last type, i'd argue, is good procrastination.

that's the "absent-minded professor," who forgets to shave, or eat, or even perhaps look where he's going while he's thinking about some interesting question. his mind is absent from the everyday world because it's hard at work in another.

that's the sense in which the most impressive people i know are all procrastinators. they're type-c procrastinators: they put off working on small stuff to work on big stuff.

what's "small stuff?" roughly, work that has zero chance of being mentioned in your obituary. it's hard to say at the time what will turn out to be your best work (will it be your magnum opus on sumerian temple architecture, or the detective thriller you wrote under a pseudonym?), but there's a whole class of tasks you can safely rule out: shaving, doing your laundry, cleaning the house, writing thank-you notes — anything that might be called an errand.

good procrastination is avoiding errands to do real work.

good in a sense, at least. the people who want you to do the errands won't think it's good. but you probably have to annoy them if you want to get anything done. the mildest seeming people, if they want to do real work, all have a certain degree of ruthlessness when it comes to avoiding errands.

some errands, like replying to letters, go away if you ignore them (perhaps taking friends with them). others, like mowing the lawn, or filing tax returns, only get worse if you put them off. in principle it shouldn't work to put off the second kind of errand. you're going to have to do whatever it is eventually. why not (as past-due notices are always saying) do it now?

the reason it pays to put off even those errands is that real work needs two things errands don't: big chunks of time, and the right mood. if you get inspired by some project, it can be a net win to blow off everything you were supposed to do for the next few days to work on it. yes, those errands may cost you more time when you finally get around to them. but if you get a lot done during those few days, you will be net more productive.

in fact, it may not be a difference in degree, but a difference in kind. there may be types of work that can only be done in long, uninterrupted stretches, when inspiration hits, rather than dutifully in scheduled little slices. empirically it seems to be so. when i think of the people i know who've done great things, i don't imagine them dutifully crossing items off to-do lists. i imagine them sneaking off to work on some new idea.

conversely, forcing someone to perform errands synchronously is bound to limit their productivity. the cost of an interruption is not just the time it takes, but that it breaks the time on either side in half. you probably only have to interrupt someone a couple times a day before they're unable to work on hard problems at all.

of course, the main reason people find it difficult to work on a particular problem is that they don't enjoy it. when you're young, especially, you often find yourself working on stuff you don't really like-- because it seems impressive, for example, or because you've been assigned to work on it. most grad students are stuck working on big problems they don't really like, and grad school is thus synonymous with procrastination.

but even when you like what you're working on, it's easier to get yourself to work on small problems than big ones. why? why is it so hard to work on big problems? one reason is that you may not get any reward in the foreseeable future. if you work on something you can finish in a day or two, you can expect to have a nice feeling of accomplishment fairly soon. if the reward is indefinitely far in the future, it seems less real.

another reason people don't work on big projects is, ironically, fear of wasting time. what if they fail? then all the time they spent on it will be wasted. (in fact it probably won't be, because work on hard projects almost always leads somewhere.)

but the trouble with big problems can't be just that they promise no immediate reward and might cause you to waste a lot of time. if that were all, they'd be no worse than going to visit your in-laws. there's more to it than that. big problems are *terrifying*. there's an almost physical pain in facing them. it's like having a vacuum cleaner hooked up to your imagination. all your initial ideas get sucked out immediately, and you don't have any more, and yet the vacuum cleaner is still sucking.

you can't look a big problem too directly in the eye. you have to approach it somewhat obliquely. but you have to adjust the angle just right: you have to be facing the big problem directly enough that you catch some of the excitement radiating from it, but not so much that it paralyzes you. you can tighten the angle once you get going, just as a sailboat can sail closer to the wind once it gets underway.

if you want to work on big things, you seem to have to trick yourself into doing it. you have to work on small things that could grow into big things, or work on successively larger things, or split the moral load with collaborators. it's not a sign of weakness to depend on such tricks. the very best work has been done this way.

when i talk to people who've managed to make themselves work on big things, i find that all blow off errands, and all feel guilty about it. i don't think they should feel guilty. there's more to do than anyone could. so someone doing the best work they can is inevitably going to leave a lot of errands undone. it seems a mistake to feel bad about that.

i think the way to "solve" the problem of procrastination is to let delight pull you instead of making a to-do list push you. work on an ambitious project you really enjoy, and sail as close to the wind as you can, and you'll leave the right things undone.

thanks to trevor blackwell, jessica livingston, and robert morris for reading drafts of this.

the web is turning writing into a conversation. twenty years ago, writers wrote and readers read. the web lets readers respond, and increasingly they do—in comment threads, on forums, and in their own blog posts.

many who respond to something disagree with it. that's to be expected. agreeing tends to motivate people less than disagreeing. and when you agree there's less to say. you could expand on something the author said, but he has probably already explored the most interesting implications. when you disagree you're entering territory he may not have explored.

the result is there's a lot more disagreeing going on, especially measured by the word. that doesn't mean people are getting angrier. the structural change in the way we communicate is enough to account for it. but though it's not anger that's driving the increase in disagreement, there's a danger that the increase in disagreement will make people angrier. particularly online, where it's easy to say things you'd never say face to face.

if we're all going to be disagreeing more, we should be careful to do it well. what does it mean to disagree well? most readers can tell the difference between mere name-calling and a carefully reasoned refutation, but i think it would help to put names on the intermediate stages. so here's an attempt at a disagreement hierarchy:

dh0. name-calling.

this is the lowest form of disagreement, and probably also the most common. we've all seen comments like this:

> u r a fag!!!!!!!!!!

but it's important to realize that more articulate name-calling has just as little weight. a comment like

> the author is a self-important dilettante.

is really nothing more than a pretentious version of "u r a fag."

dh1. ad hominem.

an ad hominem attack is not quite as weak as mere name-calling. it might actually carry some weight. for example, if a senator wrote an article saying senators' salaries should be increased, one could respond:

> of course he would say that. he's a senator.

this wouldn't refute the author's argument, but it may at least be relevant to the case. it's still a very weak form of disagreement, though. if there's something wrong with the senator's argument, you should say what it is; and if there isn't, what difference does it make that he's a senator?

saying that an author lacks the authority to write about a topic is a variant of ad hominem—and a particularly useless sort, because good ideas often come from outsiders. the question is whether the author is correct or not. if his lack of authority caused him to make mistakes, point those out. and if it didn't, it's not a problem.

dh2. responding to tone.

the next level up we start to see responses to the writing, rather than the writer. the lowest form of these is to disagree with the author's tone. e.g.

the force of a refutation depends on what you refute. the most powerful form of disagreement is to refute someone's central point.

even as high as dh5 we still sometimes see deliberate dishonesty, as when someone picks out minor points of an argument and refutes those. sometimes the spirit in which this is done makes it more of a sophisticated form of ad hominem than actual refutation. for example, correcting someone's grammar, or harping on minor mistakes in names or numbers. unless the opposing argument actually depends on such things, the only purpose of correcting them is to discredit one's opponent.

truly refuting something requires one to refute its central point, or at least one of them. and that means one has to commit explicitly to what the central point is. so a truly effective refutation would look like:

> the author's main point seems to be x. as he says:
>
> > <quotation>
>
> but this is wrong for the following reasons...

the quotation you point out as mistaken need not be the actual statement of the author's main point. it's enough to refute something it depends upon.

what it means

now we have a way of classifying forms of disagreement. what good is it? one thing the disagreement hierarchy *doesn't* give us is a way of picking a winner. dh levels merely describe the form of a statement, not whether it's correct. a dh6 response could still be completely mistaken.

but while dh levels don't set a lower bound on the convincingness of a reply, they do set an upper bound. a dh6 response might be unconvincing, but a dh2 or lower response is always unconvincing.

the most obvious advantage of classifying the forms of disagreement is that it will help people to evaluate what they read. in particular, it will help them to see through intellectually dishonest arguments. an eloquent speaker or writer can give the impression of vanquishing an opponent merely by using forceful words. in fact that is probably the defining quality of a demagogue. by giving names to the different forms of disagreement, we give critical readers a pin for popping such balloons.

such labels may help writers too. most intellectual dishonesty is unintentional. someone arguing against the tone of something he disagrees with may believe he's really saying something. zooming out and seeing his current position on the disagreement hierarchy may inspire him to try moving up to counterargument or refutation.

but the greatest benefit of disagreeing well is not just that it will make conversations better, but that it will make the people who have them happier. if you study conversations, you find there is a lot more meanness down in dh1 than up in dh6. you don't have to be mean when you have a real point to make. in fact, you don't want to. if you have something real to say, being mean just gets in the way.

if moving up the disagreement hierarchy makes people less mean, that will make most of them happier. most people don't really enjoy being mean; they do it because they can't help it.

thanks to trevor blackwell and jessica livingston for reading drafts of this.

related:

i finally realized today why politics and religion yield such uniquely useless discussions.

as a rule, any mention of religion on an online forum degenerates into a religious argument. why? why does this happen with religion and not with javascript or baking or other topics people talk about on forums?

what's different about religion is that people don't feel they need to have any particular expertise to have opinions about it. all they need is strongly held beliefs, and anyone can have those. no thread about javascript will grow as fast as one about religion, because people feel they have to be over some threshold of expertise to post comments about that. but on religion everyone's an expert.

then it struck me: this is the problem with politics too. politics, like religion, is a topic where there's no threshold of expertise for expressing an opinion. all you need is strong convictions.

do religion and politics have something in common that explains this similarity? one possible explanation is that they deal with questions that have no definite answers, so there's no back pressure on people's opinions. since no one can be proven wrong, every opinion is equally valid, and sensing this, everyone lets fly with theirs.

but this isn't true. there are certainly some political questions that have definite answers, like how much a new government policy will cost. but the more precise political questions suffer the same fate as the vaguer ones.

i think what religion and politics have in common is that they become part of people's identity, and people can never have a fruitful argument about something that's part of their identity. by definition they're partisan.

which topics engage people's identity depends on the people, not the topic. for example, a discussion about a battle that included citizens of one or more of the countries involved would probably degenerate into a political argument. but a discussion today about a battle that took place in the bronze age probably wouldn't. no one would know what side to be on. so it's not politics that's the source of the trouble, but identity. when people say a discussion has degenerated into a religious war, what they really mean is that it has started to be driven mostly by people's identities. [1]

because the point at which this happens depends on the people rather than the topic, it's a mistake to conclude that because a question tends to provoke religious wars, it must have no answer. for example, the question of the relative merits of programming languages often degenerates into a religious war, because so many programmers identify as x programmers or y programmers. this sometimes leads people to conclude the question must be unanswerable—that all languages are equally good. obviously that's false: anything else people make can be well or badly designed; why should this be uniquely impossible for programming languages? and indeed, you can have a fruitful discussion about the relative merits of programming languages, so long as you exclude people who respond from identity.

more generally, you can have a fruitful discussion about a topic only if it doesn't engage the identities of any of the participants. what makes politics and religion such minefields is that they engage so many people's identities. but you could in principle have a useful conversation about them with some people. and there are other topics that might seem harmless, like the relative merits of ford and chevy pickup trucks, that you couldn't safely talk about with others.

the most intriguing thing about this theory, if it's right, is that it explains not merely which kinds of discussions to avoid, but how to have better ideas. if people can't think clearly about anything that has bec-

"...the mere consciousness of an engagement will sometimes worry a whole day."

– charles dickens

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one reason programmers dislike meetings so much is that they're on a different type of schedule from other people. meetings cost them more.

there are two types of schedule, which i'll call the manager's schedule and the maker's schedule. the manager's schedule is for bosses. it's embodied in the traditional appointment book, with each day cut into one hour intervals. you can block off several hours for a single task if you need to, but by default you change what you're doing every hour.

when you use time that way, it's merely a practical problem to meet with someone. find an open slot in your schedule, book them, and you're done.

most powerful people are on the manager's schedule. it's the schedule of command. but there's another way of using time that's common among people who make things, like programmers and writers. they generally prefer to use time in units of half a day at least. you can't write or program well in units of an hour. that's barely enough time to get started.

when you're operating on the maker's schedule, meetings are a disaster. a single meeting can blow a whole afternoon, by breaking it into two pieces each too small to do anything hard in. plus you have to remember to go to the meeting. that's no problem for someone on the manager's schedule. there's always something coming on the next hour; the only question is what. but when someone on the maker's schedule has a meeting, they have to think about it.

for someone on the maker's schedule, having a meeting is like throwing an exception. it doesn't merely cause you to switch from one task to another; it changes the mode in which you work.

i find one meeting can sometimes affect a whole day. a meeting commonly blows at least half a day, by breaking up a morning or afternoon. but in addition there's sometimes a cascading effect. if i know the afternoon is going to be broken up, i'm slightly less likely to start something ambitious in the morning. i know this may sound oversensitive, but if you're a maker, think of your own case. don't your spirits rise at the thought of having an entire day free to work, with no appointments at all? well, that means your spirits are correspondingly depressed when you don't. and ambitious projects are by definition close to the limits of your capacity. a small decrease in morale is enough to kill them off.

each type of schedule works fine by itself. problems arise when they meet. since most powerful people operate on the manager's schedule, they're in a position to make everyone resonate at their frequency if they want to. but the smarter ones restrain themselves, if they know that some of the people working for them need long chunks of time to work in.

our case is an unusual one. nearly all investors, including all vcs i know, operate on the manager's schedule. but y combinator runs on the maker's schedule. rtm and trevor and i do because we always have, and jessica does too, mostly, because she's gotten into sync with us.

i wouldn't be surprised if there start to be more companies like us. i suspect founders may increasingly be able to resist, or at least postpone, turning into managers, just as a few decades ago they started to be able to resist switching from jeans to suits.

how do we manage to advise so many startups on the maker's schedule? by using the classic device for simulating the manager's schedule within the maker's: office hours. several times a week i set aside a